

1(a)	X = (large / permanent) vacuole ; Y = tonoplast ; A vacuolar membrane Z = nucleus ;	3
1(b)	<i>if correct named cells are given, allow as a tissue if plural, otherwise ignore e.g.</i> I cortex cell A cortex cells I root hair cell <i>at start</i> and, xylem / vascular bundle / stele, <i>at end</i> I cell structures epidermal (tissue) / epidermis cortical (tissue) / cortex A parenchyma endodermal (tissue) / endodermis pericycle any three in correct sequence ;; <i>if incorrect tissue(s) included with a correct full sequence of three, max 1 mark only two tissues stated but are in correct sequence allow one mark three tissues stated but not in correct sequence, and no incorrect tissues, allow one mark</i>	2
1(c)	any three from: cellulose, (molecules / fibres), hydrophilic / allow adhesion of water (molecules) ; <i>idea that</i> arrangement of cellulose (fibres) gives spaces (for water flow) ; AW e.g. many gaps freely permeable / not impermeable, (to water) ; A <i>idea of</i> few obstacles to, water, movement / flow e.g. (cell walls) not selective / allow continuous movement of water / allow water to move at a high rate / allow all water to pass through / (offer) path of least resistance <i>ref. to</i> hydrogen bonding ; <i>in correct context</i> e.g. cellulose molecules form hydrogen bonds with water molecules I hydrogen bonds between cellulose molecules hemicellulose / pectin, hydrophilic ;	3

Question	Answer	Marks
1(d)(i)	rough ER site of, protein / enzyme, synthesis <u>and</u> smooth ER site of, lipid / sorgoleone, synthesis ; A protein modification A lipid storage (barley plant), higher proportion of rough ER / lower proportion of smooth ER or (sorghum plant), higher proportion of smooth ER / lower proportion of rough ER ;	2

Question	Answer	Marks						
1(d)(ii)	<i>student X to max 3</i> (mechanism could be) exocytosis ; R diffusion / facilitated diffusion <i>see ecf below</i> <i>plus max two details from:</i> for, bulk transport / larger quantities ; <i>idea that</i> may be too large to exit cell (by other mechanisms) ; A are large molecules presence of, (small) Golgi bodies ; A presence of Golgi body <i>ref. to</i> vesicles as, Golgi / secretory, vesicles ; A description e.g. packaging into vesicles vesicles, move to / fuse with, cell (surface) membrane ; ecf if, diffusion / facilitated diffusion, stated as the mechanism: move down a. concentration / diffusion, gradient ; A move from high to low concentration <i>student Y to max 3 check one of three different options (A enzymes for acid phosphatases and vice versa)</i> option 1 enzymes, facilitated diffusion <u>and</u> sorgoleone, (simple / passive) diffusion ; <i>plus two marks for all information in table correct ;;</i> <table><tr><td><i>enzymes</i></td><td><i>sorgoleone</i></td></tr><tr><td>hydrophilic / are not hydrophobic / polar / AW</td><td>hydrophobic / not hydrophilic / non-polar / AW</td></tr><tr><td>pass through, transport / membrane / carrier, proteins AW A channel protein</td><td>cross hydrophobic core / between fatty acid tails / phospholipid bilayer AW</td></tr></table> <i>if two marks not gained, one mark for a correct row or correct column</i>	<i>enzymes</i>	<i>sorgoleone</i>	hydrophilic / are not hydrophobic / polar / AW	hydrophobic / not hydrophilic / non-polar / AW	pass through, transport / membrane / carrier, proteins AW A channel protein	cross hydrophobic core / between fatty acid tails / phospholipid bilayer AW	4
<i>enzymes</i>	<i>sorgoleone</i>							
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pass through, transport / membrane / carrier, proteins AW A channel protein	cross hydrophobic core / between fatty acid tails / phospholipid bilayer AW							

Question	Answer	Marks
1(d)(ii)	option 2 sorgoleone exocytosis and enzyme facilitated diffusion ; <i>plus</i> one additional point about exocytosis ; detail from enzyme column (shaded dark grey) ; option 3 enzymes exocytosis and sorgoleone diffusion ; <i>plus</i> one additional point about exocytosis ; sorgoleone column (shaded light grey) ;	